ST. JOSEPH'S COLLEGE OF ENGINEERING

(An Autonomous Institution) • OMR, Chennai - 119

LAB RECORD

Professional English Lab Integrated

CLASS CODE: EN25C09

STUDENT DETAILS

Student Name:	Jaya Suriyan S N V
Roll No:	26CS265
Date:	
Subject Name:	Professional English Lab Integrated

DEPARTMENT OF HUMANITIES & SCIENCES • AFFILIATED TO ANNA UNIVERSITY • CHENNAI

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English Lab Record | EN25C09 | Jaya Suriyan S N V

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Impressive Personality: Sridhar Vembu

Founder and technology entrepreneur

Full name	Sridhar Vembu
Known for	Co-founding the software company that evolved from AdventNet into Zoho Corporation
Education	B.Tech., Electrical Engineering - IIT Madras (1989); Ph.D. - Princeton University (1994)
Early career	Worked as a systems design engineer at Qualcomm before co-founding AdventNet in 1996

BIOGRAPHICAL NOTE

Sridhar Vembu studied electrical engineering at IIT Madras and later completed his Ph.D. at Princeton University. After working in wireless systems engineering, he joined with co-founders to build AdventNet in 1996. The company later evolved into Zoho Corporation. His career is notable for building a global software business through a long-term, product-focused approach rather than depending on venture-capital funding. He has also supported alternative pathways into technology careers through in-house training initiatives that place ability, learning and hard work above credentials alone.

QUALITIES THAT STAND OUT

Independent Thinking

He chose a bootstrapped path and focused on building products and capability for the long term.

Learning & Engineering

His path from electrical engineering to wireless systems and software shows the value of strong fundamentals and continuous learning.

Talent Development

Zoho's training initiatives demonstrate his belief that practical skill and effort can create opportunities beyond conventional credentials.

WHY HE INSPIRES ME

Sridhar Vembu inspires me because his journey connects technology, self-learning and independent problem solving. My own goal is to build a strong base in C and Java, strengthen data structures and algorithms, and grow as a software developer. His example reminds me that a strong career is not built only by collecting certificates; it also comes from understanding fundamentals, creating useful software and continuing to learn. As I start building small projects on GitHub, I want to follow the same habit of learning by building.

MY TAKEAWAY: Build strong fundamentals. Learn continuously. Create useful technology. Let practical ability and consistent work speak for themselves.

Jaya Suriyan S N V

First-Year B.E. Computer Science and Engineering Student

Degree	B.E. Computer Science and Engineering (Batch 2026 - 2030)
College	St. Joseph's College of Engineering, Chennai
Location	Tirunelveli, Tamil Nadu, India
GitHub	snvjayasuriyan027
LinkedIn	Jaya Suriyan S N V

ABOUT ME

I am a Computer Science and Engineering student with a strong interest in technology, programming and software development. I am currently building my skills in C, Java, problem solving and computer science fundamentals. I enjoy learning new technologies, working on coding challenges and developing practical solutions, and I want to keep improving my technical and communication skills through projects, internships and teamwork.

SKILLS

- C programming
- Java
- Problem solving
- Basic Git and GitHub

CURRENTLY LEARNING

- Data Structures and Algorithms
- Java
- Software development
- Professional English

ENGLISH SNAPSHOT

Reading speed test: **358 words per minute** with **100% comprehension**. Cambridge Interchange Fifth Edition Level 2 digital workbook: **143 of 240 activities completed** with a **93% average score**.

GOAL

To continuously improve my programming skills, build useful projects and grow as a software developer, while becoming a clear and confident communicator in English.

SHORT-TERM PLAN

- Practise C and Java regularly and strengthen data structures and algorithms.
- Upload small practice projects to GitHub and keep the profile active.
- Complete the remaining Cambridge workbook units and keep practising reading, writing and speaking.

SWOC Analysis

Growing as a software developer with strong English communication

STRENGTHS

• Basic knowledge of C and Java with real interest in problem solving.

• Fast reader: 358 WPM with 100% comprehension in the reading test.

• Steady English practice: 143 of 240 Cambridge workbook activities done at a 93% average.

• Active learning habit, with GitHub and LinkedIn profiles already created.

WEAKNESSES

• Programming knowledge is still at the beginner stage.

• Practical project experience is limited so far.

• Data structures and algorithms are still being learned.

• The professional network and GitHub activity are just starting.

OPPORTUNITIES

• Build small projects in C and Java and publish them on GitHub.

• Practise coding challenges to improve problem solving.

• Look for internships, projects and group work to gain experience.

• Finish the remaining Cambridge units and use Write and Improve and Speak and Improve regularly.

CHALLENGES

• Balancing college subjects with self-study in the first year.

• Keeping up with fast-changing technology.

• Staying consistent without losing motivation.

• Standing out among many students with similar goals.

DEVELOPMENT PRIORITIES

My first priority is to become confident in C and Java and to build a strong base in data structures and algorithms. Next, I will turn what I learn into small GitHub projects so that my skills are visible. Alongside this, regular English practice will help me explain technical ideas clearly in classrooms, interviews and teamwork.

The Science Behind Solar Panels

A Short Scientific Story

Meera: “Good morning, sir. I am Meera from the Bright Future Science Channel. We are here to learn about your solar panel project.”

Dr. Arun, the scientist: “Welcome, Meera. Shall I begin with how a solar panel works?”

Meera: “Yes, please, sir.”

Dr. Arun: “A solar panel is made of many solar cells, and each cell is made of silicon, a semiconductor. Sunlight is made of tiny packets of energy called photons. When photons fall on the cell, they knock electrons loose from the silicon atoms.”

Meera: “And what happens to those loose electrons, sir?”

Dr. Arun: “The cell has two layers, called n-type and p-type silicon. Where they meet, an electric field is formed, and it pushes the free electrons in one direction. This flow of electrons is an electric current. In short:”

Sunlight (photons) + silicon cell → flow of electrons (direct current)

Dr. Arun: “The panel produces direct current, so we connect it to an inverter, which converts it into the alternating current used in our homes.”

Meera: “Is all the sunlight converted into electricity?”

Dr. Arun: “No. Most modern panels convert only about 20 percent of the sunlight into electricity, so researchers are working to improve this. Do you understand what I am trying to say?”

Meera: “Yes, sir. I was amazed by how simple sunlight can become electricity. Thank you for your wonderful explanation and also for your project.”

Dr. Arun: “Thank you, Meera.”

Meera: “Thank you, sir, and thank you to everyone watching this show.”

Coherence of the Story

The Science Behind Solar Panels

The story follows a logical sequence of ideas through an interview between Meera and the scientist Dr. Arun. First, Meera introduces herself and explains the purpose of her visit. Next, she asks how a solar panel works, and Dr. Arun explains that sunlight is made of photons, which knock electrons loose from the silicon in a solar cell. The conversation then moves to what happens to these electrons: the n-type and p-type layers create an electric field that pushes them in one direction, producing an electric current. The explanation continues with the inverter, which converts the direct current into alternating current for use in homes. Meera then asks whether all the sunlight is converted, and Dr. Arun answers that modern panels convert only about 20 percent, so research is still going on. Finally, Meera thanks the scientist for his explanation and concludes the interview.

Thus, the story maintains coherence because each question leads naturally to the next, the ideas are connected in a meaningful order from the working principle to its practical use and limits, and the conclusion relates to the main topic.

Summary of the Story

The Science Behind Solar Panels

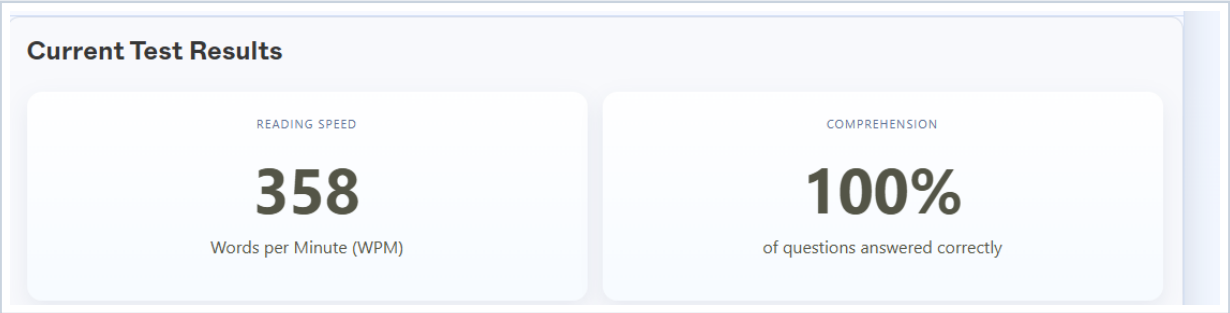
In this short interview, Meera from the Bright Future Science Channel meets the scientist Dr. Arun to learn about his solar panel project. Dr. Arun explains that a solar panel is made of silicon solar cells. When sunlight, which is made of photons, falls on a cell, it knocks electrons loose from the silicon atoms. The n-type and p-type layers of the cell create an electric field that pushes these electrons in one direction, and this flow is an electric current. Since the panel gives direct current, an inverter converts it into alternating current for homes. Dr. Arun adds that modern panels convert only about 20 percent of sunlight into electricity, so research is continuing. Meera thanks him and ends the show.

Glossary of Technical Terms

Term	Meaning
Solar cell	A device that changes sunlight directly into electricity.
Silicon	A semiconductor material used to make solar cells.
Semiconductor	A material that conducts electricity better than an insulator but not as well as a metal.
Photon	A tiny packet of light energy.
Electron	A negatively charged particle whose flow forms an electric current.
n-type and p-type silicon	Two layers of silicon treated differently; where they meet, an electric field is formed.
Direct current (DC)	Electric current that flows in one direction only.
Alternating current (AC)	Electric current that regularly changes its direction; it is the type used in homes.
Inverter	A device that converts direct current into alternating current.

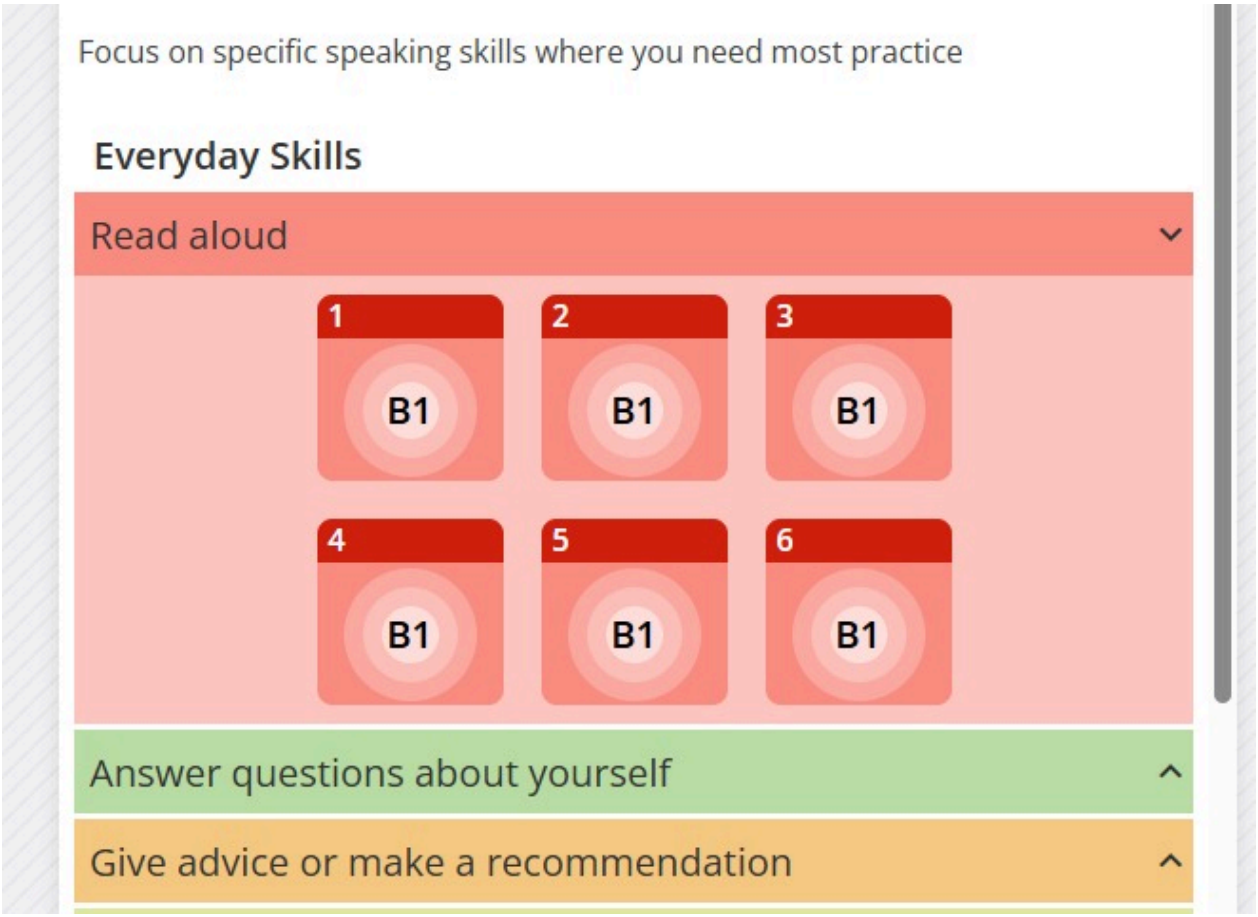
Reading Speed Test Result

Speed reading test



Reading speed: 358 words per minute, with 100% of the comprehension questions answered correctly.

SECTION 05 | SPEAK AND IMPROVE REPORT



NPTEL Summary

Enhancing Soft Skills and Personality | IIT Kanpur | Prof. T. Ravichandran

Week	Lecture	Key ideas
1	**Introduction 1: Highlights of Developing Soft Skills**	Soft skills such as communication, listening and responsibility decide how well knowledge is used. Improvement begins with a deliberate decision to change.
2	**Introduction 2: Further Highlights of Developing Soft Skills**	Self-awareness, preparation and ethical conduct matter. Observe your own habits, choose precise words and check that the listener has understood.
3	**Mindset 1: Definition and Types**	A mindset is the set of beliefs through which we interpret events. Limiting beliefs hold us back, while a positive outlook encourages us to attempt small, achievable steps.
4	**Mindset 2: Learning Mindsets**	A fixed mindset treats mistakes as proof of inability; a growth mindset treats them as feedback about the method. Progress needs several attempts and honest review.
5	**Mindset 3: Secrets of Developing Growth Mindsets**	Break big goals into small tasks, reflect on results and use the word “yet” for unfinished skills. Evidence comes from repeated, purposeful practice.
6	**Managing Time 1: Importance and Perceptions of Time**	Time is limited, so first observe where it actually goes. A realistic timetable includes interruptions, breaks and room to adjust.
7	**Managing Time 2: Using Time Efficiently**	Set priorities, estimate effort, limit distractions and fix a clear output for each session. Review the results to see which tasks need more time.

WHAT I TAKE FROM THE COURSE

Technical skill alone is not enough; I also need to explain my ideas clearly, listen to others and learn from feedback. As a programming student, I can treat compiler errors as feedback instead of failure, plan my study time in focused blocks for C, Java and data structures, and practise explaining my code in simple words.

Ideation: Smart Study Planner

A simple project idea (planned, not yet built)

IDEAS CONSIDERED

Idea	What it does	Decision
Quiz game in C	Asks multiple-choice questions and shows the score.	Good practice, but limited scope
Expense tracker	Records daily spending and totals it.	Useful, but less related to my studies
Smart Study Planner	Stores study tasks with subject and deadline, and lists what is due first.	**Chosen**: solves a real student problem

PROBLEM

First-year students handle many subjects, assignments and records at once, and it is easy to forget a deadline.

IDEA

A simple Java console application where a student adds study tasks, sees pending tasks sorted by due date and marks tasks as completed.

PLANNED FEATURES

- Add a task with subject name, description and due date.
- Show all pending tasks sorted by the nearest deadline.
- Mark a task as completed and view completed tasks.
- Save tasks in a file so they are available the next day.

TOOLS AND SKILLS USED

Java (classes, objects, collections, file handling), a sorting step from data structures, and Git and GitHub for saving the project.

STEPS TO BUILD

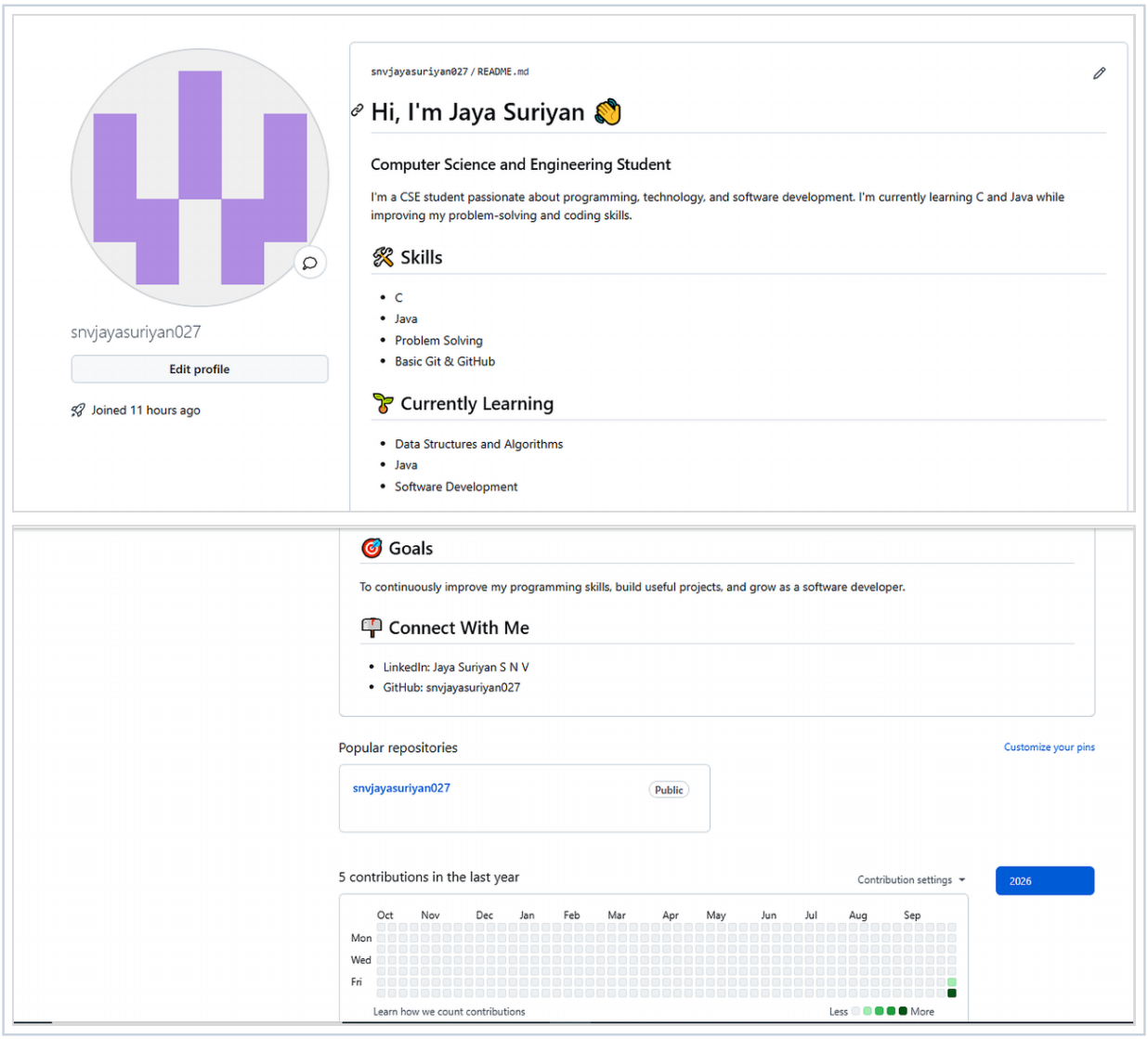
1. Design the Task class and the menu.
2. Write the add, view and complete functions.
3. Add file saving and loading.
4. Test with sample tasks and upload the code to GitHub.

FUTURE SCOPE

A simple graphical or mobile version with reminders, which would also help me practise software development beyond the console.

GitHub Profile

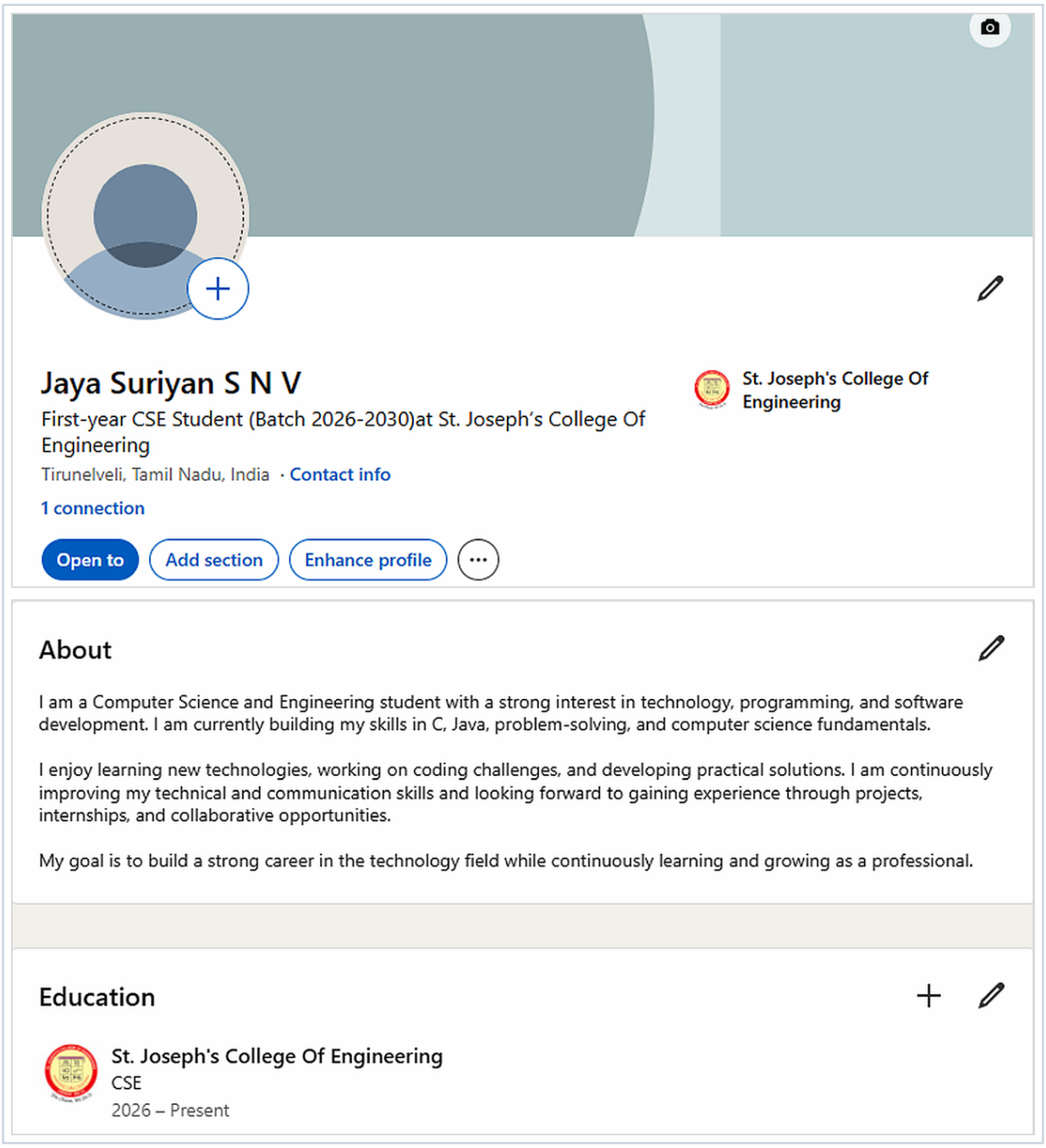
GitHub username: snvjayasuriyan027



Screenshot of my GitHub profile and README.

LinkedIn Profile

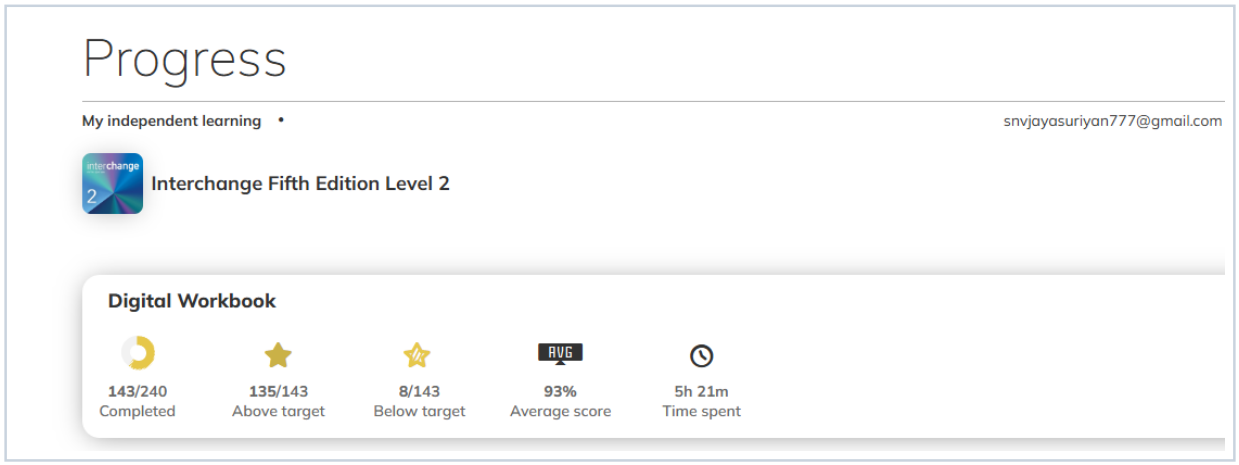
LinkedIn: Jaya Suriyan S N V



Screenshot of my LinkedIn profile (header, About and Education).

Cambridge Interchange Progress

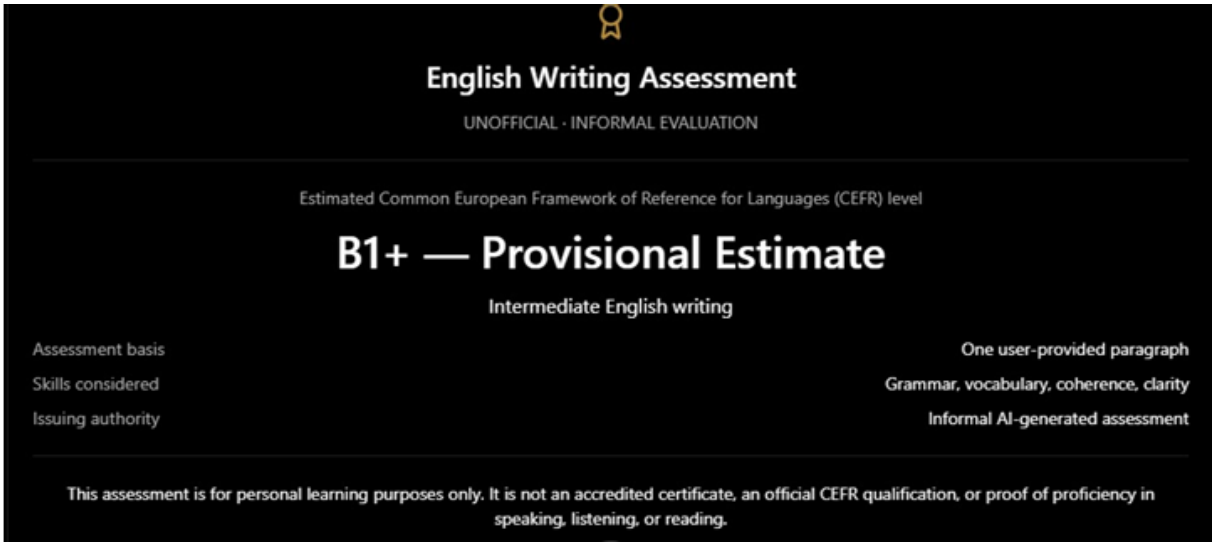
Interchange Fifth Edition Level 2 | Digital Workbook



Digital workbook: 143 of 240 completed, 135 above target, 8 below target, 93% average score, 5 h 21 m spent.

CEFR Level Estimate

English Writing Assessment (informal)



Screenshot of the informal English writing assessment: B1+ provisional estimate. It is an unofficial, AI-generated evaluation for personal learning.

Language Level of the Story

Informal CEFR Estimate of the Text

B1+ (borderline B2)

Provisional estimate of this story's text only

Item	Observation
Assessment basis	The story “The Science Behind Solar Panels” (interview format, about 260 words).
Grammar	Mostly simple and present tenses; correct use of passive forms (“is made of”), relative ideas and “when” clauses. Few complex structures.
Vocabulary	Everyday words with technical terms such as semiconductor, photons, electric field, inverter, alternating current.
Coherence	Clear order of ideas from introduction to explanation to limits to conclusion; questions lead naturally to answers.
Clarity	Short, direct sentences that a general audience can follow easily.
Issuing authority	Informal AI-generated assessment

This estimate is for personal learning purposes only. It describes the language of this text, not an individual writer's proficiency, and it is not an accredited certificate or an official CEFR qualification.

VOCABULARY WORKSHEET

Complete Answer Key

Answers organised by section and part

Name: Jaya Suriyan S N V

Roll No: 26CS265

1. Jobs and Recruitment

Part A — Fill in the blanks

1. do / do
2. in charge
3. shifts
4. headhunters
5. CV

Part B — Matching

1. Flexitime — B
2. Teleworking — C
3. Psychometric test — E
4. Shortlist — D
5. Referees — A

Part C — Sentence correction

1. I am in charge of the design department.
2. He goes to work by train every morning.
3. She is very responsible in her daily tasks.

2. Employment and Pay

Part A — Fill in the blanks

1. graduate
2. unskilled
3. commission
4. share options
5. fat cats

Part B — Matching

1. Skilled — D
2. Good — E
3. Computer — A
4. Self-starter / motivated — B
5. Fringe — C

Part C — Short answers

1. Salary: usually a fixed amount paid monthly or annually. Wages: commonly calculated by hours, days, or work completed.
2. Severance package: compensation when employment ends. Bonus: extra payment, often for performance or achieving targets.

3. Workplace Issues

Part A — Fill in the blanks

1. white-collar
2. strike
3. downsized and delayered
4. redundant
5. glass ceiling

Part B — Matching

1. Repetitive Strain Injury — B
2. Passive smoking — D
3. Sexual harassment — A
4. Equal opportunities — C
5. Whistleblowing — E

Part C — Short answers

1. Dismissed: employment ends because of conduct, performance, or another employment-related reason. Redundant: the job is no longer required.
2. An open-plan office is a shared open working area without separate enclosed offices for each employee.

4. Company Leadership

Part A — Fill in the blanks

1. non-executive
2. executives
3. entrepreneur
4. tycoons
5. privatized

Part B — Matching

1. SME — A
2. Corporate culture — D
3. Enterprise zone — E
4. Head up — B
5. Boardroom — C

Part C — Sentence correction

1. He heads up our marketing department.
2. We need to buy an e-commerce website/platform for our new online shop.

5. Business Ownership and Sectors

Part A — Fill in the blanks

1. sole trader
2. PLC
3. charities
4. beta
5. recall

Part B — Sector classification

1. Aerospace — M
2. Catering — S
3. Pharmaceuticals — M
4. Property / real estate — S
5. Steel production — M

Part C — Short answers

1. CADCAM: computer-aided design and computer-aided manufacturing.
2. A focus group is a small group of people brought together to discuss a product, service, or idea and provide feedback.

6. Production

Part A — Fill in the blanks

1. obsolete
2. patent
3. capacity
4. inputs
5. just-in-time

Part B — Matching

1. R&D — B
2. Royalties — A
3. Craft industry — C
4. Work-in-progress — E
5. Outsourcing — D

Part C — Short answer

1. Just-in-Time (JIT) production receives materials when needed, reducing inventory, storage costs, waste, and money tied up in stock.

7. Quality and Competition

Part A — Fill in the blanks

1. conformance
2. benchmarking
3. clients
4. market leader
5. cut-throat

Part B — Matching

1. Kaizen — B
2. Mystery shopper — C
3. Market segment — E
4. Key players — D
5. BPR — A

Part C — Short answer

1. A market leader has a leading position or the largest share in a market. A challenger is a competitor trying to gain market share and compete with the leader.

8. Marketing Mix

Part A — Fill in the blanks

1. place
2. consumer durables
3. generic
4. loss leaders
5. niche

Part B — Matching

1. FMCG — A
2. Own-brand — B
3. Upmarket — D
4. Trade up — C
5. Price war — E

Part C — Short answer

1. The Four Ps are Product, Price, Place, and Promotion.

9. Distribution, Retail, Advertising and Financial Terms

Part A — Financial fill-ins

1. turnover
2. fixed
3. cash cows
4. accounts receivable
5. auditors

Part B — Financial matching

1. COGS — A
2. Goodwill — C
3. Depreciation — B
4. In the red — E
5. Window dressing — D

Part C — Distribution and retail fill-ins

1. the trade
2. chain
3. junk
4. campaign
5. bricks

Part D — Matching

1. Cold call — C
2. Franchise — E
3. Billboard — A
4. Loyalty card — D
5. B2B — B

10. Meetings

Part A — Fill in the blanks

1. chair
2. agenda
3. compromise
4. hedging
5. deadlock

Part B — Matching

1. AGM — C
2. Minutes — E
3. Consensus — A
4. Action points — B
5. Hedging — D

Part C — Short answers

1. The chairperson prepares or reviews the agenda, arranges the meeting, keeps discussion on track, and ensures participants can contribute.
2. Consensus means general agreement among participants; voting selects an option by counting votes, even if some disagree.

11. Presentations

Part A — Fill in the blanks

1. visual aids
2. rapport
3. introduction
4. body language
5. audience

Part B — Matching

1. Visual aids — C
2. Body language — B
3. Rapport — D
4. Q&A session — E
5. Handouts — A

Part C — Sentence correction

1. During my presentation, I will show you some visual aids.
2. Let’s open the floor for some questions from the audience.

12. Negotiations

Part A — Fill in the blanks

1. win-win
2. trade-offs
3. deadlock
4. contract
5. mediator

Part B — Matching

1. Probing — A
2. Counter-proposal — D
3. Deadlock — B
4. Concession — E
5. Mediator — C

Part C — Short answers

1. Hard bargaining focuses on securing the best result for one side; a win-win approach seeks an outcome that benefits both parties.
2. Checking the final contract confirms that agreed terms, prices, responsibilities, deadlines, and conditions are accurately recorded and understood.

13. Financial Markets

Part A — Fill in the blanks

1. stock market
2. credit
3. derivatives
4. bull
5. blue

Part B — Matching

1. Forex — B
2. Bourse — C
3. NASDAQ — A
4. FTSE — E
5. Bear market — D

Part C — Short answers

1. A futures contract commits the parties to transact at agreed terms on a specified future date. An option gives its buyer the right, but not the obligation, to transact under specified terms.
2. “The City” refers to London’s financial district and “Wall Street” to New York’s financial centre; both names are also used for their financial industries.

14. Economics

Part A — Fill in the blanks

1. finance
2. trade surplus
3. Gross Domestic Product (GDP)
4. recession
5. slump

Part B — Matching

1. Inflation rate — C
2. Trade deficit — D
3. Stagflation — E
4. Trough — B
5. Stagnation — A

Part C — Sentence correction

1. The country has a high rate of inflation.
2. When output reaches its lowest point in the business cycle, it reaches a trough.

15. Business Crime and Ethics

Part A — Fill in the blanks

1. insider dealing
2. price fixing
3. embezzlement
4. money laundering
5. ethical investing

Part B — Matching

1. Embezzlement — D
2. Counterfeiting — A
3. Bribery — B
4. Code of ethics — E
5. Sweatshop labour — C

Part C — Short answers

1. Chinese walls are information barriers within an organisation designed to prevent improper sharing of confidential information between departments.
2. Social audits assess a company's social and ethical practices, such as worker treatment, community impact, and environmental responsibility.

16. Time and Stress

Part A — Fill in the blanks

1. timescale / timeframe
2. delay
3. challenging
4. burnout
5. rat race

Part B — Matching

1. Downtime — D
2. Prioritize — E
3. Perfectionism — C
4. Downshifting — A
5. Stress-induced — B

Part C — Short answers

1. Perfectionism can waste time, increase stress, delay completion, and make prioritising tasks difficult.
2. Downshifters often value well-being, personal time, and work-life balance more than salary or job status alone.

17. Leadership and Culture

Part A — Fill in the blanks

1. subordinates
2. visionary
3. consultation
4. empowerment
5. macho

Part B — Matching

1. Authoritarian — B
2. Decentralized — E
3. Deference — D
4. Corporate culture — C
5. Canteen culture — A

Part C — Sentence correction

1. The manager is very remote, so she is completely unapproachable by all staff.
2. We operate a bottom-up, consultative approach in our department.

18. Business Etiquette

Part A — Fill in the blanks

1. first names
2. dress-down Friday
3. presenteeism
4. cordless phone
5. toll-free

Part B — Matching

1. Surname — B
2. Smart casual — E
3. Punctuality — D
4. Pager — C
5. Videoconferencing — A

Part C — Short answers

1. In many English-speaking workplaces, colleagues use first names relatively quickly, although expectations vary by person and context.
2. Cultures differ in how strictly they follow punctuality and schedules; learn and respect the expectations of the people you work with.

19. Meetings

Part A — Fill in the blanks

1. AGM
2. agenda
3. minutes
4. open
5. views

Part B — Matching

1. Chairperson — C
2. Agenda — E
3. Minutes — D
4. Action points — B
5. Venue — A

Part C — Short answers

1. A chairperson guides discussion, keeps the meeting on track, manages time, gives participants a chance to speak, and summarizes decisions.
2. An agenda sent in advance lets participants prepare, gather information, and understand the meeting's topics and objectives.

20. Discussion and Negotiation

Part A — Fill in the blanks

1. hedging
2. win-win
3. interests
4. trade-offs
5. deadlock

Part B — Matching

1. Hedging — D
2. Consensus — C
3. Probing — E
4. Counter-proposal — B
5. Mediator — A

Part C — Sentence correction

1. We reached a deadlock, so we needed a mediator to help us reach a win-win compromise that satisfied both parties.
2. During the meeting, he used too little hedging, so his point was extremely direct and clear.

21. Presentation Preparation

Part A — Fill in the blanks

1. launch
2. time
3. delivery
4. rapport
5. aids

Part B — Matching

1. Preparation — D
2. Rapport — A
3. Body language — B
4. Visual aids — C
5. Q&A session — E

Part C — Short answers

1. Do: welcome the audience, introduce yourself and the topic, explain the purpose, and outline the presentation. Don't: use a long introduction, irrelevant jokes, excessive detail, or read every word from slides.
2. Stay calm, listen, clarify the question if needed, respond politely with facts, and admit honestly if you do not know an answer.

22. Negotiation Strategy

Part A — Fill in the blanks

1. position
2. walk-away
3. agreement
4. bargaining
5. deal

Part B — Matching

1. Bargaining — C
2. Concession — D
3. Counter-proposal — A
4. Negotiating style — E
5. Deal — B

Part C — Sentence correction

1. During the bargaining phase, we made concessions and negotiated the terms before signing the contract.
2. His negotiating style was confrontational, meaning he shouted down every proposal made by the other side.

St. JOSEPH'S COLLEGE OF ENGINEERING

(An Autonomous Institution) | OMR, Chennai - 119

EN25C09: Professional English - I (LAB INTEGRATED)

MODEL EXAM I - PRACTICE WORKSHEET

Name: Jaya Suriyan S N V **Department:** B.E. Computer Science and Engineering **Year/Sem:** I / I

PART - A (15 X 1 = 15 Marks)

1. Fill in the blank with the correct form of the verb given in bracket:
The automated system _____ (monitor) the temperature continuously to ensure safe operation.

Answer: monitors

2. Fill in the blank with the correct past form of the verb given in bracket:
The engineers _____ (record) the vibration readings during yesterday's trial.

Answer: recorded

3. Complete the sentence with the appropriate comparative form of the adjective in bracket:
The newly developed composite material is significantly _____ (strong) than the conventional material while maintaining a lower weight.

Answer: stronger

4. Fill in the blank with an appropriate modal auxiliary verb: (must/can/might)
Engineers _____ follow safety regulations while operating heavy machinery.

Answer: must

5. Fill in the blank with a suitable modal auxiliary verb expressing capability: (can, may, might)
Modern sensors _____ detect even minor changes in temperature.

Answer: can

6. Combine the sentences into a single coherent sentence using a suitable relative pronoun:
(a) The laboratory developed a new testing device. (b) The device measures material strength with high precision.

Answer: The laboratory developed a new testing device that measures material strength with high precision.

7. Complete the sentence using the correct form of the verb in bracket (First Conditional - If Clause):
If the temperature _____ (rise) above the permissible limit, the cooling system will activate automatically.

Answer: rises

8. Complete the sentence using the correct form of the verb in bracket (Third Conditional - Past Unreal Conditional):
If the technician had inspected the circuit earlier, the equipment _____ (not / fail) during operation.

Answer: would not have failed

9. Combine the following sentence using an infinitive of purpose (to + verb):
The mechanics installed dual damper springs. They wanted to isolate chassis resonance.

Answer: The mechanics installed dual damper springs to isolate chassis resonance.

10. Fill in the blank with the gerund form of the verb in bracket after the preposition:
The research team reduced energy consumption by _____ (optimize) the motor control algorithm.

Answer: optimizing

11. Convert the following active voice into the Impersonal Passive Voice:
The lab assistant poured 50 mL of hydrochloric acid into the titration flask.

Answer: 50 mL of hydrochloric acid was poured into the titration flask.

12. Fill in the blank with the appropriate adverbial form derived from the adjective in bracket:
The robotic arm moves _____ (smooth) to ensure accurate positioning of the component.

Answer: smoothly

13. Rewrite the direct into indirect speech:
The project supervisor asked the engineer, "Did you calibrate the pressure sensor before testing?"

Answer: The project supervisor asked the engineer whether he had calibrated the pressure sensor before testing.

14. Identify the incorrect noun form and rewrite the sentence correctly:
The laboratory purchased several new equipments for conducting material testing.

Answer: The incorrect noun form is "equipments". "Equipment" is an uncountable noun and has no plural form. Correct sentence: The laboratory purchased several new pieces of equipment for conducting material testing.

15. Fill in the blank using the correct irregular comparative form (better, worse, farther/further, less, more):
Because the heat sink had poor thermal paste contact, heat dissipation was _____ (bad) than expected.

Answer: worse

PART - B (10 X 2 = 20 Marks)

1. Rewrite the following active operational and laboratory description in the Simple Past Tense:
The engineer calibrates the pressure sensor and verifies its accuracy.

Answer: The engineer calibrated the pressure sensor and verified its accuracy.

2. Draft two comparative sentences comparing a laptop and a desktop computer in terms of portability and screen size.

Answer (Portability): A laptop is more portable than a desktop computer because it is lighter and has a built-in battery.

Answer (Screen Size): A desktop computer usually has a larger screen than a laptop, so it gives a wider viewing area.

3. Combine the technical statements into a single cohesive sentence using a defining or non-defining relative clause:
The engineer designed a new cooling system. It reduces the temperature of industrial machines.

Answer: The engineer designed a new cooling system, which reduces the temperature of industrial machines.

4. Complete the following conditional sentences logically:

(a) If the battery voltage drops below the required level, **the device will switch off automatically to protect the circuit.**

(b) If the technician had checked the connections earlier, **the short circuit would not have occurred.**

5. Convert the following active sentence into the Impersonal Passive Voice suitable for formal engineering reports:

The laboratory assistant heated the solution to 80°C before testing.

Answer: The solution was heated to 80°C before testing.

6. Rewrite the following direct engineering query into Reported (Indirect) Speech:

The project manager asked the technician, "Have you inspected the pressure valves before starting the compressor?"

Answer: The project manager asked the technician whether he had inspected the pressure valves before starting the compressor.

7. Identify and correct the grammatical errors involving mass/uncountable nouns in the following sentences:

(a) The engineer received many feedbacks on the new design. **Correction:** The engineer received a lot of feedback on the new design. ("Feedback" is uncountable, so it has no plural and cannot take "many".)

(b) The students received several useful advices from their teacher. **Correction:** The students received several pieces of useful advice from their teacher. ("Advice" is uncountable.)

8. Rewrite the sentences using an Infinitive of Purpose or a Gerund construction:

(a) The technician wears safety gloves to protect his hands.

(b) The cooling fan operates continuously to prevent the motor from overheating. (or: The cooling fan prevents the motor from overheating by operating continuously.)

9. Fill in the blanks with the correct adverbial forms of the bracketed words:

(a) The machine operates **efficiently** to reduce energy consumption.

(b) The technician checked the electrical connections **carefully** before starting the machine.

10. Insert suitable modal auxiliary verbs to match the specified functional meaning:

(a) The robotic arm **can** (capability) lift heavy components with precision.

(b) Visitors **must not** (strict prohibition) enter the testing area without authorization.

PART - C (5 X 13 = 65 Marks)

Question 26

26. a) Write a set of eight clear, imperative safety instructions to promote safe and responsible use of the internet and protect users from cyber threats. (13 Marks)

Answer:

Safe and Responsible Use of the Internet: Safety Instructions

1. **Create strong passwords.** Use a long mix of letters, numbers and symbols, and never use the same password for different accounts.
2. **Enable two-factor authentication.** Turn it on for email, banking and social media accounts to add an extra layer of protection.
3. **Do not click on suspicious links or attachments.** Check the sender's address carefully before opening any email, message or file.
4. **Never share personal information.** Keep your OTP, passwords, bank details and Aadhaar number private, even if someone claims to be from your bank.
5. **Update your software regularly.** Install the latest updates for your operating system, browser and antivirus to fix security weaknesses.
6. **Use only secure websites.** Look for "https" and the padlock symbol before entering any login or payment details.
7. **Avoid public Wi-Fi for private work.** Do not log in to banking or other sensitive accounts on open networks; use a trusted network or a VPN.
8. **Think before you post and report cyber crime.** Be respectful online, do not share unverified content, and report cyberbullying or fraud to the cyber crime portal (cybercrime.gov.in) or helpline 1930.

Question 27

27. a) Write a descriptive essay of 250 words about "A Traditional art form in your region and how modern initiatives can preserve its cultural significance." Discuss its historical origins, traditional practices, and ways to promote and sustain it among younger generations. (13 Marks)

Workspace / Outline Area:

Selected Question: 27.a

Introduction: Kolam, the daily floor art of Tamil Nadu, and what it stands for.

Body Paragraph 1: Historical origins and traditional practices (rice flour, pulli kolam, Margazhi and Pongal).

Body Paragraph 2: Modern threats and initiatives (competitions, workshops, social media, apps, exhibitions).

Conclusion: Kolam as heritage, science and devotion; the need to pass it to the young.

Answer:

Kolam: The Living Art at the Doorstep of Tamil Nadu

In Tamil Nadu, the day begins with a quiet and beautiful ritual. Before sunrise, women sprinkle water outside their homes and draw delicate patterns with rice flour. This traditional art form is called kolam, one of the most cherished symbols of Tamil culture.

The origins of kolam go back many centuries. Ancient Tamil literature mentions decorated courtyards, and the tradition has passed from mother to daughter. Rice flour is used deliberately, because it feeds ants and birds, which reflects harmony with nature. Kolam is also believed to welcome the goddess Lakshmi and bring prosperity.

Traditionally, a kolam is drawn freehand with a pinch of flour held between the fingers. Many designs begin with a grid of dots called pulli kolam, and unbroken lines are looped around them. This needs patience, memory and a sense of geometry. During Margazhi and Pongal, the patterns grow larger and more colourful.

Today, busy lifestyles, apartment living and ready-made stickers threaten this heritage. Modern initiatives can protect it. Schools and colleges can organise kolam competitions and workshops, and social media can share tutorials. Mobile apps can teach the mathematics behind the patterns, which will attract engineering students. Cultural groups can support festivals and exhibitions that honour skilled artists, and training camps can help young people to learn from them.

In conclusion, kolam is not just decoration; it is heritage, science and devotion drawn in a single line. If the younger generation learns it with pride, this art will brighten our doorsteps for years to come.

Question 28

28. b) Draft a formal Letter of Request to the Principal of your college seeking permission to participate in a technical workshop on Artificial Intelligence and Emerging Technologies organised by a reputed institution. Mention the objectives of participation, the expected learning outcomes, and the arrangements required. (13 Marks)

Workspace / Format Outline:

Selected Question: 28.b

Sender's Details / Date: Jaya Suriyan S N V, B.E. CSE, First Year; 21 October 2026

Recipient's Details: The Principal, St. Joseph's College of Engineering, OMR, Chennai - 119

Subject: Request for permission to attend a technical workshop on AI and Emerging Technologies

Salutation: Respected Madam/Sir,

Content / Body: Purpose, objectives, learning outcomes, arrangements required, closing request.

Sign-off: Yours obediently / Yours faithfully, name and signature

Answer:

From
Jaya Suriyan S N V,
B.E. Computer Science and Engineering (First Year),
St. Joseph's College of Engineering,
OMR, Chennai - 119.

Date: 21 October 2026

To
The Principal,
St. Joseph's College of Engineering,
OMR, Chennai - 119.

Subject: Request for permission to participate in a technical workshop on Artificial Intelligence and Emerging Technologies

Respected Madam/Sir,

I am a first-year B.E. Computer Science and Engineering student of your college. I would like to inform you that [Name of the Institution] is organising a technical workshop on "Artificial Intelligence and Emerging Technologies" on [Date of the Workshop]. I kindly request your permission to participate in this workshop.

Objectives of participation: My main objectives are to understand the basic concepts of Artificial Intelligence and machine learning, to learn how these technologies are applied in real-life fields such as cloud computing, healthcare and automation, and to gain practical exposure through hands-on sessions that go beyond my classroom syllabus.

Expected learning outcomes: After attending the workshop, I expect to gain clear knowledge of current technological trends, improve my technical and problem-solving skills, and be able to apply what I have learnt in my academic projects. I will also share the knowledge with my classmates and prepare a short report on the workshop for the department.

Arrangements required: I request you to grant me on-duty permission for the day(s) of the workshop and to issue a permission or recommendation letter for the registration. I will make my own travel arrangements and the registration fee, if any, will be paid by me. I will also take care to catch up on all the classes and assignments I may miss.

I assure you that I will represent the college with discipline and will submit the participation certificate after the workshop. I shall be grateful if you kindly grant me permission.

Thanking you,

Yours obediently,
Jaya Suriyan S N V
B.E. CSE, First Year

Enclosure: Workshop brochure / invitation

Question 29

29. a) Compose an analytical essay (250-300 words) comparing the advantages and disadvantages of Artificial Intelligence-Based Assessment versus Traditional Teacher-Based Assessment in engineering education. Evaluate assessment accuracy, feedback time, fairness, and the role of human judgement. (13 Marks)

Workspace / Comparative Matrix and Outline:

Selected Question: 29.a

Criteria 1 vs. Criteria 2: AI-based assessment vs. teacher-based assessment

Pros and Cons Breakdown: Accuracy - AI is consistent on objective tasks, teachers understand reasoning. Feedback time - AI is instant, teachers are slow. Fairness - AI has no personal bias but may have data bias, teachers may be subjective but consider context. Human judgement - strongest in teachers.

Drafting Notes: Introduction, accuracy, feedback time, fairness, human judgement, conclusion (blended approach).

Answer:

AI-Based Assessment versus Traditional Teacher-Based Assessment in Engineering Education

Assessment is the process by which an institution measures how well students have learnt. With the rise of Artificial Intelligence, many engineering colleges are now asking whether automated evaluation can replace the traditional method in which teachers check papers and award marks.

In terms of accuracy, AI-based systems are very strong in objective tasks such as multiple-choice tests, coding exercises and numerical problems. They apply the same rules every time and do not make mistakes caused by tiredness. A teacher, however, understands the reasoning behind a student's answer and can give partial credit for a correct method even when the final result is wrong.

Feedback time is another major difference. An AI tool can evaluate hundreds of answer scripts within minutes and give instant feedback, which helps students correct their mistakes quickly. Manual evaluation often takes days or even weeks, which delays improvement.

On the question of fairness, AI is free from mood and personal favouritism, so every student is judged by the same standard. Yet an AI system can be unfair if it is trained on limited data, and it may misjudge creative or unusual answers. Teachers can be influenced by handwriting or personal impressions, but they are able to consider a student's effort and background.

The role of human judgement is where the traditional method is strongest. Teachers can assess creativity, communication, teamwork and ethical thinking, which machines still find difficult to measure. They also motivate and guide students personally.

In conclusion, neither method is perfect. AI offers speed, consistency and instant feedback, while teachers offer understanding, flexibility and care. The best solution for engineering education is a blended approach, in which AI handles routine evaluation and teachers review complex answers and make the final decisions.

Question 30

30. b) Draft a personal email to your school friend describing your experience attending the orientation programme for first-year engineering students at your college. Highlight the activities conducted, the introduction to academic facilities and college regulations, and how the programme helped you understand college life and prepare for your academic journey. (13 Marks)

Workspace / Email Structure:

Selected Question: 30.b

To: Beno

Subject Line: My first-year orientation at college

Salutation: Dear Beno,

Opening / Check-in: Ask about his well-being and studies; say why I am writing.

Core Experience / Details: Activities, academic facilities, rules and regulations, benefits of the programme.

Closing: Invite him to reply and share his experiences; best wishes.

Answer:

To: beno@example.com
Subject: My first-year orientation at college

Dear Beno,

Hope you are doing well and enjoying your college life. I am fine, and I am writing to tell you about the orientation programme for first-year engineering students that I attended at St. Joseph's College of Engineering.

The programme began with a welcome address by the Principal and the Heads of the Departments, who spoke about the vision of the college and the importance of discipline and hard work. Then our seniors and faculty members organised ice-breaking games and a short interaction session. These activities helped me to overcome my nervousness and make new friends in my class.

We were also taken on a guided tour of the campus. We visited the library, the laboratories, the computer centre, the seminar halls and the sports facilities. The faculty explained how to use the library resources and the lab equipment safely. They also explained the college regulations, such as the attendance requirement, the dress code, the examination system and the rules for using the college transport and the campus.

The orientation really helped me to understand what college life is like. I now know what is expected of me, and I feel more confident and prepared for my academic journey. I have decided to attend my classes regularly and to make the best use of the facilities available here.

Please write to me about your college and how your first days went. Let us also meet during the next holidays and share our experiences.

Take care and best wishes for your studies.

Yours affectionately,
Jaya Suriyan S N V